

EEP2 – Week 7

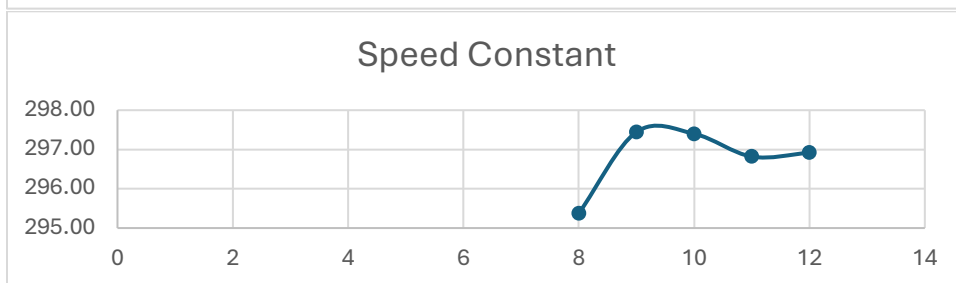
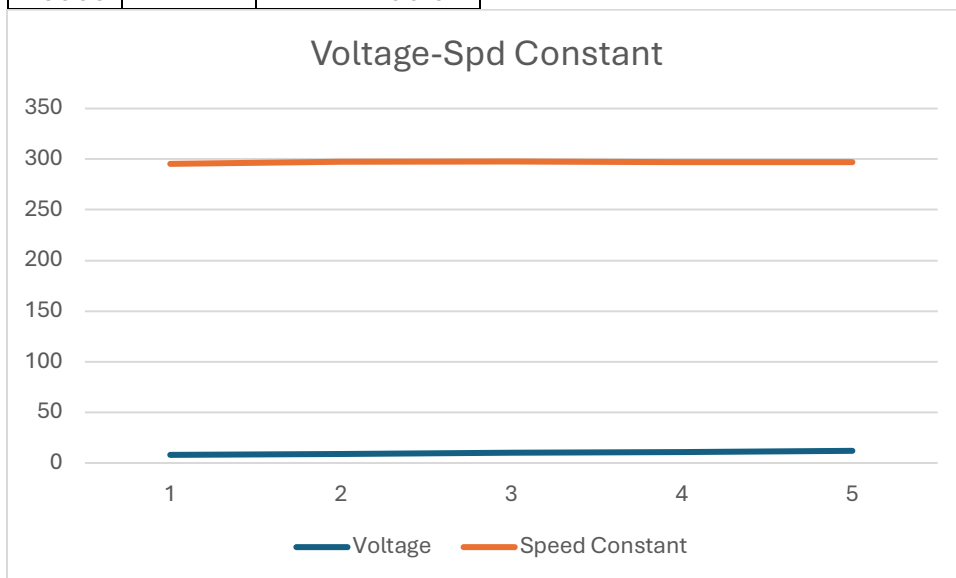
7/3/2025 – E4A-06-03

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Lab

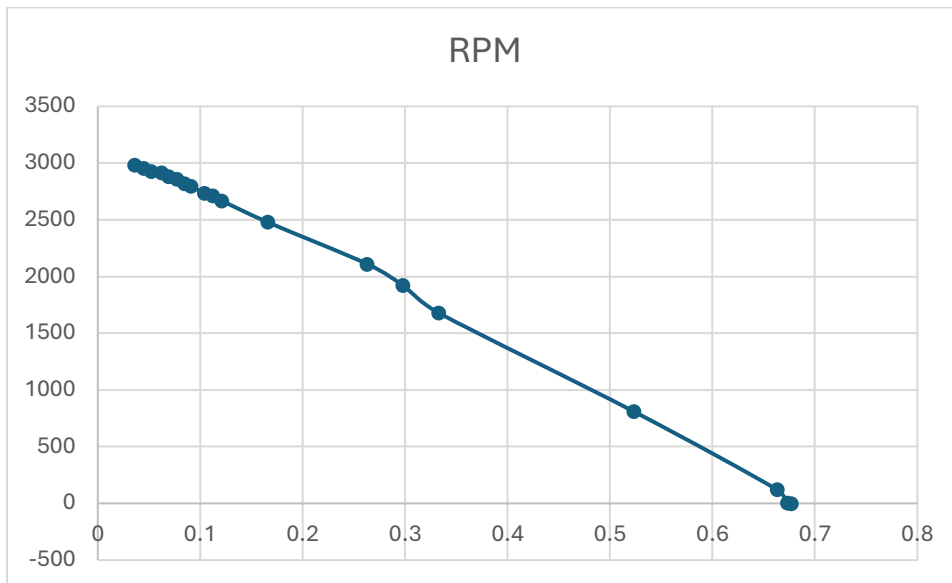
Activity #1a

RPM	Voltage	Speed Constant
2363	8	295.38
2677	9	297.44
2974	10	297.40
3265	11	296.82
3563	12	296.92



Activity #1b

Current	RPM
0.036	2982
0.045	2952
0.052	2928
0.062	2914
0.069	2882
0.077	2856
0.085	2817
0.091	2794
0.104	2736
0.112	2713
0.121	2668
0.166	2478
0.263	2110
0.298	1920
0.333	1678
0.523	811
0.663	121
0.673	5
0.677	0



Find K_e :

$$2\pi \frac{RPM}{60} = \frac{V_a}{K_e} - \frac{R_a}{K_e} I_a$$
$$2\pi \frac{2982}{60} = \frac{10}{K_e} - 0$$
$$K_e = 0.0320 Nm/A$$

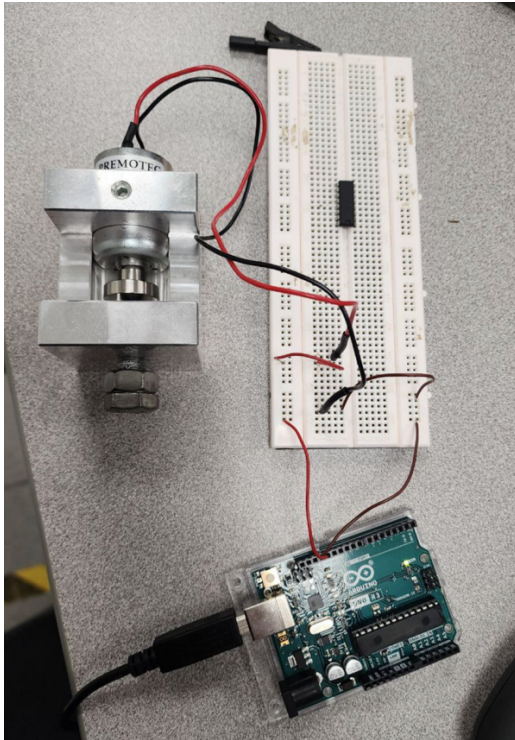
Find R_a :

Gradient of line: -0.0002119

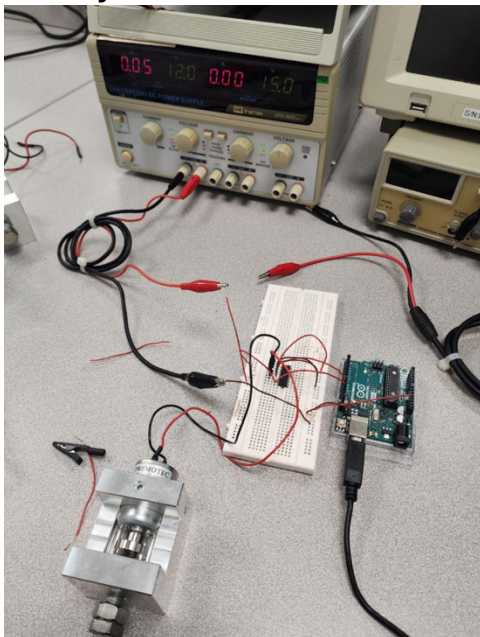
Find K_t :

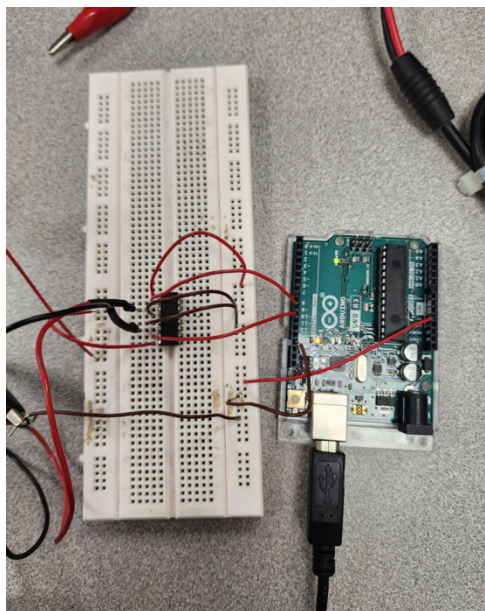
$$K_t = K_e \times \sqrt{R_a}$$
$$K_t = 0.0320 \times \sqrt{0.0002119} = 0.0004658$$

Activity #2a



Activity #2b





```
#define ENA 5
#define IN1 9
#define IN2 10

void setup() {
    pinMode(ENA, OUTPUT);
    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);
}

void motorControl(int speed, bool direction) {
    analogWrite(ENA, speed);

    if (direction) {
        digitalWrite(IN1, HIGH);
        digitalWrite(IN2, LOW);
    } else {
        digitalWrite(IN1, LOW);
        digitalWrite(IN2, HIGH);
    }
}

void loop() {
    Serial.print("50% forward");
    motorControl(128, true); // 50% forward
    delay(3000);

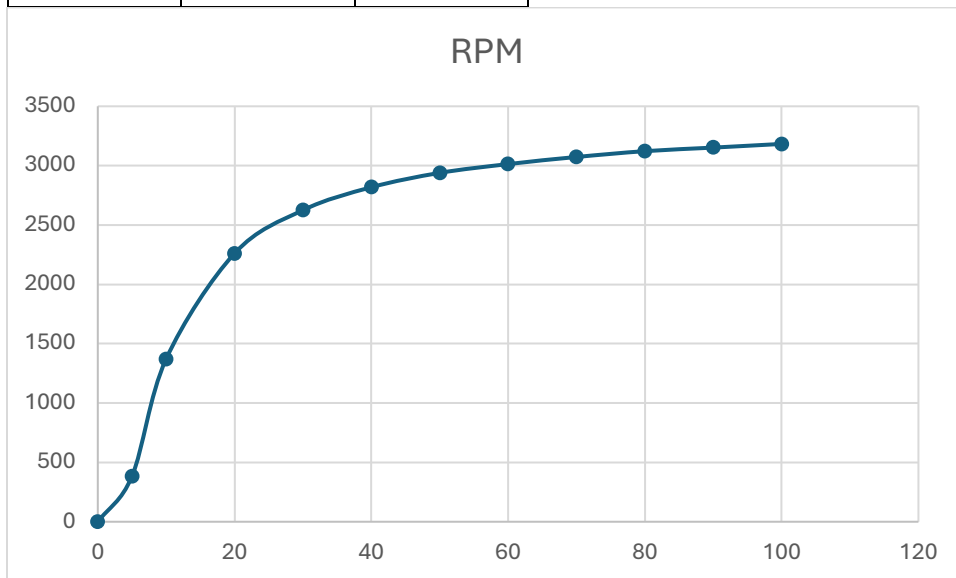
    Serial.print("100% forward");
    motorControl(255, true); // 100% forward
    delay(3000);

    Serial.print("50% reverse");
    motorControl(128, false); // 50% reverse
    delay(3000);

    Serial.print("Stop");
    motorControl(0, true); // Stop
    delay(2000);
}
```

Activity #2c

Full	Duty Cycle	RPM
255	100	3183
229.5	90	3152
204	80	3122
178.5	70	3073
153	60	3013
127.5	50	2939
102	40	2818
76.5	30	2625
51	20	2261
25.5	10	1372
12.75	5	383
0	0	0



```
#define ENA 5
#define IN1 9
#define IN2 10

int dutyCycle = 128;

void setup() {
    pinMode(ENA, OUTPUT);
    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);

    Serial.begin(9600);
    Serial.println("Enter Duty Cycle (0-255):");

    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
}

void loop() {
    analogWrite(ENA, dutyCycle);

    if (Serial.available()) {
        int newDutyCycle = Serial.parseInt();
        if (newDutyCycle >= 0 && newDutyCycle <= 255) {
            dutyCycle = newDutyCycle;
            Serial.print("New Duty Cycle: ");
            Serial.println(dutyCycle);
        }
    }

    delay(3000);
}
```